

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN SOILS

MODULE - 1

SOIL AND ITS CONTENTS

- **PROCESS OF SOIL FORMATION**
- **SOIL FORMING FACTORS**

INDIAN GEOGRAPHY

PLAN OF ACTION



1. INDIAN MONSOON
2. INDIAN SOILS
3. INDIAN NATURAL VEGETATION
4. INDIAN PHYSIOGRAPHY
 - HIMALAYAN RANGES
 - NORTHERN PLAINS
 - PENINSULAR PLATEAU
 - COASTAL PLAINS OF INDIA
 - ISLANDS OF INDIA
 - HILL RANGES OF THE PENINSULAR PLATEAU



INDIAN GEOGRAPHY

PLAN OF ACTION

5. MINERAL DISTRIBUTION IN INDIA

- IRON-ORE DISTRIBUTION
- COAL FORMATION AND DISTRIBUTION
- PETROLEUM & ITS DISTRIBUTION
- MANGANESE ORE & ITS DISTRIBUTION
- DISTRIBUTION OF GOLD & SILVER
- COPPER, NICKEL & CHROMITE
- NON-FERROUS MINERALS DISTRIBUTION
- DISTRIBUTION OF NUCLEAR MINERALS
- DISTRIBUTION OF DIAMOND & GRAPHITE
- NON-METALLIC MINERALS IN INDIA



INDIAN GEOGRAPHY

PLAN OF ACTION



6. NATURAL HAZARDS & DISASTERS

7. INDIAN DRAINAGE SYSTEM

- INDUS RIVER SYSTEM
- GANGA RIVER SYSTEM
- BRAHMAPUTRA RIVER SYSTEM
- PENINSULAR RIVER SYSTEM



SOIL AND SOIL CONTENTS



- 1** Soil is a dynamic medium in which many chemical, physical and biological activities go on constantly.
- 2** Soil is a **result of decay**. It is also the medium for growth. It is a changing and developing body.
- 3** It has **many characteristics that fluctuate with the seasons**. It may be alternatively cold and warm or dry and moist.
- 4** Biological activity is slowed or stopped if the soil becomes too cold or too dry.

SOIL AND SOIL CONTENTS



5 The soil chemistry, the amount of organic matter, the soil flora and fauna, the temperature and the moisture, all change with the seasons as well as with more extended periods of time.

6 That means, soil becomes adjusted to conditions of climate, landform and vegetation and will change internally when these controlling conditions change.



HOW DO
SOILS
FORM?

PROCESS OF SOIL FORMATION

Weathered material gets colonized by bacteria and other inferior plant bodies like mosses and lichens.



Dead remains of organisms and plants help in humus accumulation.



Minor grasses may grow; later, bushes and trees start growing through seeds brought by wind and birds.



Plant roots and burrowing animals mix up earth material.

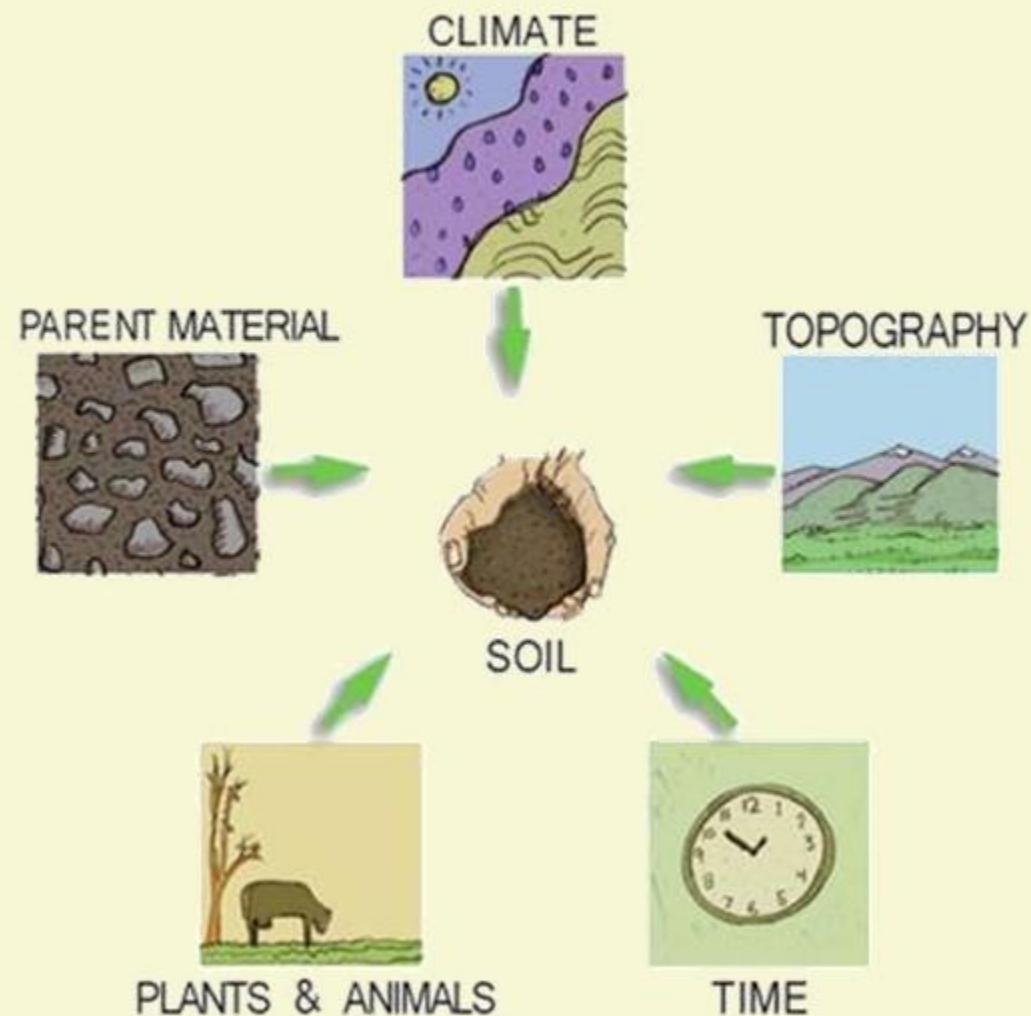


Earth material becomes porous and sponge-like to permit of air and also capacity to retain water.



Mature soil gets formed, a complex mix of mineral and organic products.

SOIL-FORMING FACTORS



Five basic factors control the formation of soils:

- i. Parental material
- ii. Topography
- iii. Climate
- iv. Biological activity
- v. Time.

In fact soil forming factors act in union and affect the action of one another.

THE CLORPT FACTOR!

$$S = f(cl, o, r, p, t)$$

SOIL-FORMING FACTORS

1

PARENT MATERIAL



1 Parent material is a **passive control factor** in soil formation. Parent materials can be any **in-situ or on-site weathered rock debris** (residual soils) or **transported deposits** (transported soils).

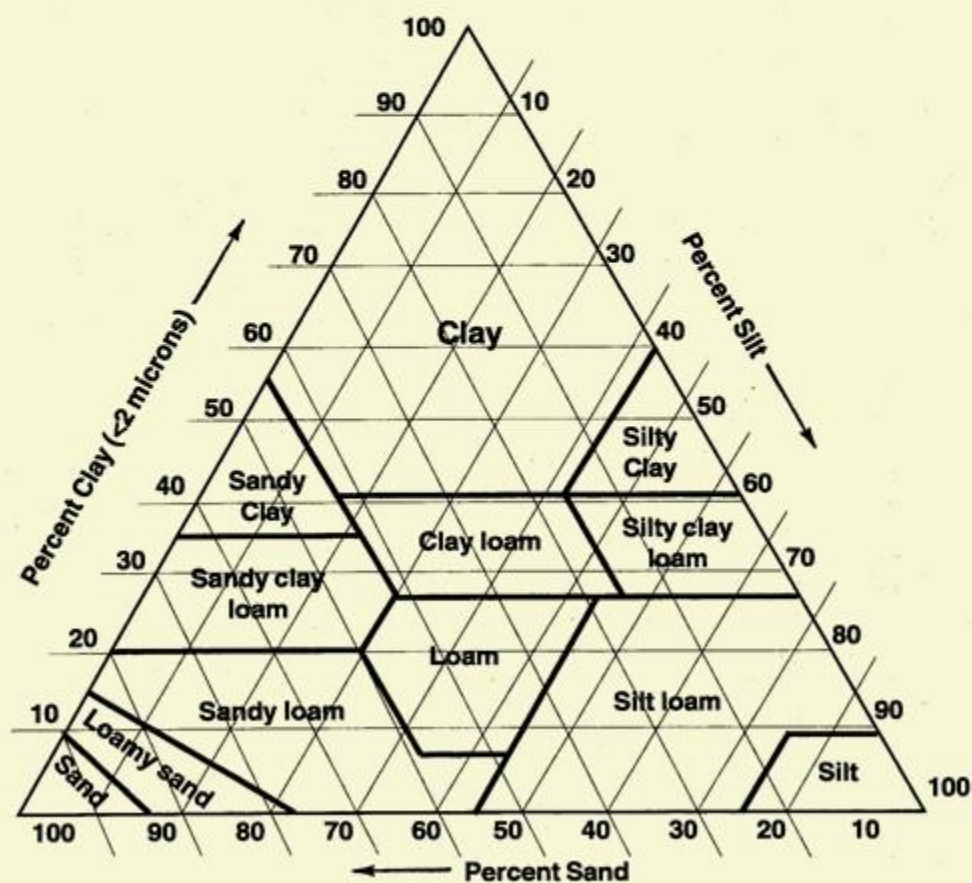
2 Soil formation depends upon the **texture** (sizes of debris) **and structure** (disposition of individual grains/particles of debris) as well as the **mineral and chemical composition** of the rock debris/deposits.

SOIL-FORMING FACTORS

1

PARENT MATERIAL

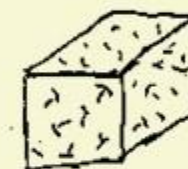
TEXTURE VS STRUCTURE



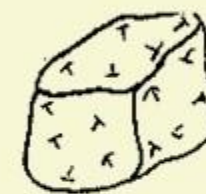
Granular



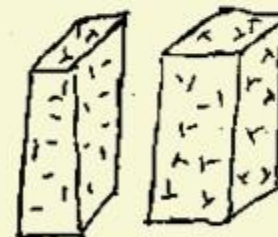
Crumb



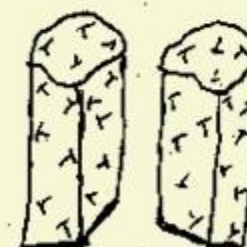
Blocky



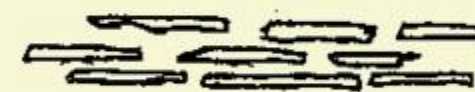
Subangular blocky



Prismatic



Columnar



Platy



Single grain/structureless

SOIL-FORMING FACTORS

1

PARENT MATERIAL



3 Nature and **rate of weathering and depth of weathering mantle** are important consideration under parent materials.

4 There may be differences in soil over similar bedrock and dissimilar bedrocks may have similar soils above them.

5 But, when soils are very young and have not matured these show strong links with the type of parent rock.

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